Robotic-assisted versus conventional minimally invasive esophagectomy: a propensity-matched cohort study

A. Example, B. Sample, C. Placeholder (fictional authors for software testing)

Methods

Patients who underwent robotic-assisted minimally invasive esophagectomy (RAMIE) or conventional minimally invasive esophagectomy (MIE) for esophageal cancer between 2015 and 2020 were identified from a prospectively maintained institutional database. The choice of approach was made by the operating surgeon.

Propensity score matching (1:1) was performed using age, sex, ASA classification, body mass index, clinical stage and neoadjuvant therapy. Smoking history and surgeon experience were not available in the database and were not included in the model.

Anastomotic leak was defined according to the Esophagectomy Complications Consensus Group (ECCG) definitions and confirmed by contrast study or endoscopy. Outcome assessors were not blinded to the surgical approach.

Follow-up began on the date of surgery. Patients with missing data on clinical stage (n = 12) were excluded from the analysis. No imputation was performed.

Results

A total of 412 patients were identified. After matching, 120 patients were included in each group. Baseline characteristics were well balanced after matching.

Anastomotic leak occurred in 14 (11.7%) patients after RAMIE and 22 (18.3%) after MIE (OR 0.59, 95% CI 0.29 to 1.21, p = 0.15).

Table 2. Postoperative outcomes in the matched cohort
Anastomotic leak 14 (11.7%) 22 (18.3%) 0.15
Pneumonia 18 (15.0%) 25 (20.8%) 0.24

Discussion

The study was not registered and no protocol or statistical analysis plan was published. Residual confounding by unmeasured factors such as frailty cannot be excluded. No sensitivity analyses for unmeasured confounding were performed.